

Cross-Coin Heterogeneity in Liquidation Cascade Dynamics: Evidence from Meme Coin Perpetual Futures

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Abstract

Exchange-reported liquidation data is unreliable, with major venues throttling feeds to one event per second. We propose a framework detecting liquidation cascades from public order book data, bypassing exchange-reported feeds. Applying it to three meme coin perpetual futures on OKX—PEPE, BOME, and FLOKI—we document significant cross-coin heterogeneity: positive cascade rates in small-cap coins (BOME 19.4%, FLOKI 18.2%) significantly exceed that of large-cap PEPE (12.5%, $p < 0.001$). OBI-Bias coupling is positive in BOME ($\rho = +0.14$, $p < 0.001$) but negative in PEPE ($\rho = -0.23$, $p < 0.001$), suggesting that market structure fundamentally alters the sign of the order book–price relationship. These findings caution against uniform risk parameters across structurally distinct assets.

JEL Classification: G10, G14, G15, G17

Keywords: liquidation cascades, perpetual futures, meme coins, order book imbalance, market microstructure, OKX, blockchain architecture

1 Introduction

Perpetual futures contracts account for over 93% of all cryptocurrency derivatives trading volume, exceeding \$90 trillion in cumulative notional volume since 2020 (Ruan and Streltsov, 2024). A defining feature of these markets is the liquidation cascade—a self-reinforcing cycle in which declining prices trigger forced closures of over-leveraged long positions, further depressing prices and triggering additional liquidations (Gudgeon et al., 2020). During extreme events, cascades can erase hundreds of millions in open interest within minutes. Understanding their dynamics is therefore central to risk management, exchange design, and financial stability in crypto markets.

Yet exchange-reported liquidation data is systematically unreliable: since mid-2021, major venues have throttled their liquidation feeds to one event per second, rendering the data “wildly underreported” (Lunde, 2024). Liquidation cascades are among the most economically significant events in perpetual futures markets, yet their empirical study has been constrained by the very exchanges that host them.

We address this opacity by detecting liquidation cascades from public order book and open interest data—without relying on exchange-reported feeds. We monitor three microstructure indicators—order book imbalance (OBI), price bias, and open interest delta—through OKX’s public WebSocket channels. Flagged events are labeled via the trading system’s exit logic: events triggering a trailing take-profit (price rebound $\geq 0.3\%$) are genuine cascades; events hitting the stop-loss (-1.0%) are false signals; timeouts are labeled by whether peak return exceeds 0.3% . This builds on the order imbalance methodology of (Easley et al., 2012; Koutmos and Wei, 2023), adapted to forced liquidation detection.

We apply this framework to three meme coin perpetual futures that trade on the same exchange (OKX) and share the same liquidation engine: PEPE, BOME, and FLOKI. The three coins are selected by a transparent criterion: we include all meme coin perpetual futures on OKX that meet minimum data requirements (continuous WebSocket availability and at least 200 cascade-like events during the collection period). Despite their shared “meme” label, the three coins occupy structurally distinct positions. PEPE, an

ERC-20 token on Ethereum, is the largest meme coin by market capitalization, with deep order books and ~ 12 -second block times. BOME, an SPL token on Solana, and FLOKI, deployed on both BNB Chain and Ethereum, are smaller-cap meme coins on faster blockchains (Solana at ~ 0.4 s and BNB Chain at ~ 3 s, respectively), characterized by thinner order books and higher volatility. This design holds the exchange mechanism constant while varying market structure. Recent work documents that meme coins generate significant volatility spillovers to broader crypto markets (Galati and Perdichizzi, 2025), yet their microstructure-level liquidation dynamics remain unstudied.

Data was collected continuously from June 2–5, 2026. Although spanning only 3.5 days, this period constitutes a natural experiment in compressed form: June 2–3 provided calm baseline conditions, while June 4–5 delivered a market-wide sell-off (BTC declining 13.5%, over \$1.5 billion in liquidations across exchanges). The resulting 8,454 cascade-like events capture the full spectrum from quiescence to extreme liquidation stress within a single contiguous window, making the sample unusually information-dense for studying cross-coin heterogeneity in cascade dynamics. We document two principal findings. First, the three coins exhibit substantial cross-coin heterogeneity in positive cascade rates: BOME (19.4%) and FLOKI (18.2%) significantly exceed PEPE (12.5%), with bootstrap 95% confidence intervals excluding zero for both comparisons ($p < 0.001$). The difference between BOME and FLOKI is not statistically significant ($p = 0.224$), suggesting a binary distinction between large-cap deep-book meme coins and small-cap thin-book meme coins. Second, this heterogeneity maps onto observable market structure differences: PEPE’s mean order book imbalance ($\text{OBI} = -0.70$) indicates persistent sell-side pressure; BOME exhibits a moderate skew (-0.38), while FLOKI is close to neutral (-0.12), and the exit-reason composition differs markedly across coins, with FLOKI and BOME exhibiting higher trailing-stop activation rates (9.0% and 10.2%) than PEPE (3.8%).

These findings contribute to the growing literature on cryptocurrency market microstructure (Natashekara and Sampath, 2024; Ruan and Streltsov, 2024) by providing, to our knowledge, the first direct cross-coin comparison of liquidation cascade dynamics using real-time order book data. The documented heterogeneity challenges a tacit as-

sumption in the empirical crypto literature—that assets within a given class share similar microstructural properties. For exchanges, this suggests that uniform risk parameters (e.g., margin requirements, position limits) may be mis-calibrated across coins; for researchers, it implies that findings from a single cryptocurrency may not generalize to others, even within the same asset category.

The remainder of this paper is organized as follows. Section 2 describes the OKX data infrastructure and the three-indicator detection framework. Section 3 presents our empirical results. Section 4 discusses implications and limitations.

2 Data and Detection Framework

2.1 OKX WebSocket Infrastructure

We collect real-time market data from OKX’s public WebSocket API, which provides two channels relevant to our detection task. The `open-interest` channel pushes the exchange-wide open interest for each perpetual swap at approximately 500ms intervals. The `books5` channel pushes the top five levels of the order book (bid and ask) at the same frequency. Both channels are freely accessible without authentication. Data is collected via a Python-based listener deployed on a virtual private server (Lightlayer, Hong Kong) with sub-10ms latency to OKX’s Asia-Pacific endpoint, ensuring minimal network-induced measurement error.

We compute open interest delta as a percentage change ($\Delta OI_t = (OI_t - OI_{t-1}) / OI_{t-1}$) rather than in raw units. The percentage form is unit-agnostic and directly comparable across coins, requiring no instrument-specific calibration.

2.2 Three-Indicator Detection Framework

Our detection framework monitors three market microstructure indicators in real time, each computed from the WebSocket data streams at every update tick:

Order Book Imbalance (OBI) measures the asymmetry between bid and ask depth at the top five levels of the order book: $OBI = (V_{bid} - V_{ask}) / (V_{bid} + V_{ask})$, where V_{bid}

and V_{ask} denote total quantity at the top five bid and ask levels, respectively. OBI ranges from -1 (complete ask dominance, i.e., sell-side pressure) to $+1$ (complete bid dominance). During liquidation cascades, forced selling rapidly consumes bid-side liquidity, driving OBI toward -1 .

Price Bias measures the deviation of the current mid-price from its two-minute simple moving average, expressed as a percentage. A large negative bias indicates an unusually rapid price decline—the hallmark of a liquidation-driven sell-off.

Open Interest Delta measures the percentage change in exchange-wide open interest over the most recent observation interval. While OI delta is included as an input feature to the detection model, it is not used as a standalone trigger condition—OI changes are typically small and gradual outside of extreme events.

A cascade event is flagged when any of four OR-trigger conditions is met: (i) OBI drops below a coin-specific extreme threshold (Path 1: PEPE -0.90 , BOME -0.95 , FLOKI -0.92); (ii) OBI recently breached its extreme level and is rebounding while Price Bias remains below a coin-specific level (Path 2); (iii) Price Bias drops below a coin-specific threshold (Path 3: PEPE -0.20% , BOME -0.30% , FLOKI -0.20%); or (iv) OI Delta $< -0.5\%$ AND OBI < -0.4 (Path 4). In practice, Path 4 did not trigger for any of the three meme coins during our collection period—consistent with their small contract sizes and correspondingly low OI variability—but is retained for completeness as it was the primary trigger for ETH under the system’s earlier deployment. Flagged events are subject to a 5-second cooldown plus a state-based filter (OBI change < 0.01 and Bias change < 0.001) to suppress duplicate triggers from the same cascade. The coin-specific calibration of thresholds is itself evidence of cross-coin heterogeneity: PEPE’s deeper order books permit a less stringent OBI trigger (-0.90) than BOME’s thin books (-0.95).

2.3 Event Labeling

Each flagged event is tracked for 120 seconds (the system’s maximum hold duration) and labeled according to its exit outcome within the trading system:

- **Label = 1** (genuine liquidation cascade): the event triggered a trailing take-profit (price rebounded $\geq 0.3\%$, then retraced 0.15% from the peak) *or* the event timed out but reached a peak return $> 0.3\%$ (indicating a missed exit opportunity at a genuine bounce).
- **Label = 0** (false signal): the event hit the hard stop-loss (-1.0%) *or* timed out without ever reaching 0.3% peak return.

This labeling scheme has the advantage of being grounded in the trading system’s actual exit logic rather than an arbitrary return threshold. Importantly, the trigger indicators (OBI and Price Bias) are measured at the moment of event detection, while the label is determined by the price trajectory over the subsequent 120 seconds. This temporal separation—trigger conditions evaluated at $t = 0$, label determined by outcomes at $t + 120$ s—ensures that the labeling criterion is not mechanically related to the triggering variables. The trailing take-profit condition—activation at $+0.3\%$ followed by a 0.15% retracement from the peak—provides an objective, non-discretionary criterion for identifying genuine post-cascade price rebounds.

2.4 Cross-Coin Sample

We apply the detection framework to three meme coin perpetual futures traded on OKX: PEPE-USDT-SWAP, BOME-USDT-SWAP, and FLOKI-USDT-SWAP. These three tokens are issued on three different blockchains—PEPE as an ERC-20 token on Ethereum, BOME as an SPL token on Solana, and FLOKI on BNB Chain and Ethereum—allowing us to compare liquidation dynamics across blockchain architectures while holding the exchange mechanism constant. The three coins were selected by including all meme coin perpetual futures on OKX that met minimum data requirements during the collection period (continuous WebSocket data availability, minimum 200 cascade-like events). Larger meme coins (DOGE, SHIB) are excluded: their deep, slow-moving order books resemble mainstream assets rather than the volatile meme coin phenomenon under study. Data was collected continuously over June 2–5, 2026, spanning both calm conditions and a sig-

nificant market-wide sell-off (BTC down 13.5% weekly, over \$1.5 billion in liquidations). Table 1 reports sample sizes and summary statistics.

Despite their shared “meme coin” label, the three coins occupy structurally distinct positions. PEPE (launched April 2023) is the largest Ethereum-based meme coin and has developed deep, institutional-like order books. BOME (March 2024) is a Solana-based token whose 10,131 SOL presale produced extreme early volatility and the thinnest books in our sample. FLOKI (June 2021) operates on BNB Chain and Ethereum, occupying an intermediate position.

Blockchain architecture further differentiates the three coins in ways directly relevant to liquidation dynamics. Solana’s ~ 0.4 -second block times enable near-instantaneous arbitrage between spot and perpetual markets, narrowing price dislocations faster than Ethereum’s ~ 12 -second blocks (Biais et al., 2019; Hautsch et al., 2024). This implies that liquidation-induced price deviations on BOME (Solana) should be absorbed more rapidly than on PEPE (Ethereum), producing shorter-duration but more frequent cascade events. BNB Chain’s ~ 3 -second blocks place FLOKI in an intermediate position. We emphasize that our design holds the exchange-level liquidation engine constant (all three coins trade on OKX and are subject to the same mark-price and liquidation rules), meaning that any observed cross-coin differences in cascade dynamics are attributable to market microstructure rather than institutional mechanism.

3 Results

3.1 Cross-Coin Summary Statistics

Table 1 reports summary statistics for the 8,454 cascade-like events across the three coins. Substantial cross-coin heterogeneity is immediately apparent. The positive cascade rate—the proportion of flagged events that result in a genuine liquidation cascade, as determined by the trailing-stop-based labeling rule—ranges from 12.5% for PEPE to 19.4% for BOME (with FLOKI at 18.2%), approximately a 1.6-to-one difference. PEPE’s mean order book imbalance of -0.70 indicates persistent, structural sell-side pressure. BOME’s mean OBI

(-0.38) reflects moderate sell-side skew, while FLOKI (-0.12) is close to neutral.

The OBI discriminative power—the difference in mean OBI between positive and negative events (ΔOBI)—is comparable for PEPE ($+0.29$) and BOME ($+0.26$), while FLOKI (-0.01) shows essentially no discriminative power. However, the aggregate ΔOBI masks substantial within-coin stratification. Table 2 partitions PEPE’s events by OBI zone. The extreme negative zone ($\text{OBI} < -0.95$), which accounts for 68.2% of all triggers, has a positive rate of only 9.6%—a “falling knife” zone where sell pressure is overwhelming and rebounds are rare. In contrast, the positive OBI zone ($\text{OBI} > 0.5$), representing only 5.0% of triggers, achieves a 40.8% positive rate—a genuine signal zone where buyers have returned to the order book. This bimodal structure—a deep noise zone and a shallow signal zone separated by approximately 31 percentage points in accuracy—is unique to PEPE. BOME, despite its similar ΔOBI , differs fundamentally: its mean OBI (-0.38) indicates persistent moderate selling pressure across all events, lacking PEPE’s sharp positive-signal zone. FLOKI’s OBI is close to neutral (-0.12) with minimal discriminative power. For BOME, the discriminating feature is not OBI but Price Bias (-0.61% for positive events vs. -0.77% for negative), consistent with thinner order books where price deviations, rather than depth imbalances, signal cascade outcomes.

Table 1: Cross-Coin Summary Statistics

	PEPE	BOME	FLOKI
Observed events (N)	6,281	1,142	1,031
Positive rate (%)	12.5	19.4	18.2
Mean OBI	-0.696	-0.377	-0.123
Mean Price Bias (%)	-0.157	-0.736	-0.334
Mean Peak Return (%)	0.151	0.246	0.150
OBI (label = 1)	-0.446	-0.165	-0.134
OBI (label = 0)	-0.732	-0.428	-0.120
ΔOBI (1 – 0)	$+0.286$	$+0.263$	-0.013
OBI–Bias correlation (ρ)	-0.230	$+0.135$	-0.135
Exit: Trail TP (%)	3.8	10.2	9.0
Exit: Stop Loss (%)	4.4	22.9	8.4
Exit: Timeout (%)	91.8	67.0	82.5

Table 2: OBI Stratification and Signal Content: PEPE

OBI Zone	Events (N)	% of Total	Positive Rate (%)
$> +0.5$	311	5.0	40.8
0 to $+0.5$	399	6.4	15.5
-0.5 to 0	937	14.9	14.6
-0.95 to -0.5	352	5.6	12.2
< -0.95	4282	68.2	9.6

Table 3: Cross-Exchange Validation: 5-Second REST Tick Data

	PEPE	BOME	FLOKI
<i>Spread (%)</i>			
Binance	0.0034	0.0236	0.0392
Bybit	0.0330	0.0461	0.0492
<i>Volatility (5s return std, %)</i>			
Binance	0.050	0.059	0.048
Bybit	0.051	0.061	0.048

Note: 39,600 observations per coin per exchange, 5-second REST polling. BOME and FLOKI spreads significantly exceed PEPE on both exchanges (Mann–Whitney, $p < 0.001$). PEPE and BOME volatility differences are directionally consistent with OKX data.

3.2 OBI Distribution and Market Structure

Figure 1 contrasts OBI distributions by coin and cascade outcome. PEPE is strongly left-skewed (bid-side liquidity chronically thin), yet positive events are visibly right-shifted relative to negative events ($\Delta\text{OBI} = +0.29$). FLOKI is centered near zero with low variance—its cascades appear driven by transient shocks that may resolve between our 500ms samples. BOME occupies a middle ground: moderate sell-side skew (-0.38) with higher peak returns, reflecting the violent but short-lived nature of its liquidation events.

3.3 OBI–Bias Coupling and the Blockchain Mechanism

The most striking cross-coin difference emerges in the relationship between order book imbalance and price, displayed in Figure 2. In PEPE, OBI and Price Bias exhibit a significant negative correlation ($\rho = -0.230$, $p < 0.001$): as the order book becomes more sell-dominated, the price declines. This is the canonical market microstructure relationship predicted by standard models (Easley et al., 2012; Cont, 2011)—order book pressure transmits to price movement.

In BOME, the relationship reverses sign: OBI is positively correlated with Price Bias ($\rho = +0.135$, $p < 0.001$), meaning that periods of elevated sell-side pressure tend to coincide with upward price movements—a pattern consistent with mean reversion in extreme thin-book conditions, where panic selling is rapidly met by contrarian buying. FLOKI exhibits weak negative coupling ($\rho = -0.135$, $p < 0.001$). A Fisher z -transformation confirms that all three pairwise differences are statistically significant: PEPE differs from BOME ($z = 11.47$, $p < 0.001$), PEPE from FLOKI ($z = 2.93$, $p = 0.003$), and BOME from FLOKI ($z = 6.30$, $p < 0.001$).

The sign reversal in BOME is particularly notable: whereas PEPE’s order book operates in the canonical microstructure regime—sell pressure transmits to price declines—BOME’s thin, fast-moving market exhibits the opposite pattern, where extreme sell-side imbalance attracts rapid contrarian buying, resulting in a net positive OBI–Bias correlation. This is consistent with the limits-to-arbitrage framework of (Hautsch et al., 2024): in markets with minimal settlement latency (Solana’s ~ 0.4 s blocks) and thin order books, arbitrage capital can flow in quickly to exploit temporary dislocations, inverting the usual order-book–price relationship.

This finding has implications beyond our specific application. If OBI’s informational content varies systematically across market structures, then microstructure indicators developed and validated on established, deep-book assets may not transfer to thinner, less mature markets without recalibration.

3.4 Exit Reason Composition

Figure 3 reports the composition of exit reasons by coin. The trailing take-profit activation rate—the proportion of events in which the price rebounded sufficiently to trigger profit-taking—is highest for BOME (10.2%), followed by FLOKI (9.0%), and lowest for PEPE (3.8%). This ordering is consistent with the positive rate patterns documented above and reinforces the binary distinction: small-cap, thin-book meme coins (BOME, FLOKI) experience more frequent and more pronounced post-cascade rebounds than the large-cap, deep-book PEPE.

However, BOME also exhibits the highest stop-loss rate (22.9%, versus 4.4% for PEPE and 8.4% for FLOKI). This is the risk side of the opportunity: BOME’s higher cascade frequency comes with a higher probability that a flagged event is actually a continued decline rather than a temporary liquidation dip. FLOKI’s moderate stop-loss rate (8.4%) and neutral OBI structure distinguish it from both PEPE (low risk, moderate signal) and BOME (high risk, high signal).

3.5 Statistical Significance

A bootstrap test (2,000 iterations) on pairwise positive rate differences supports a binary classification: PEPE differs significantly from BOME (+7.0 pp, 95% CI [+4.5%, +9.4%], $p < 0.001$) and FLOKI (+5.8 pp, [+3.3%, +8.3%], $p < 0.001$), while BOME and FLOKI are statistically indistinguishable (+1.2 pp, [−2.0%, +4.5%], $p = 0.224$).

4 Discussion and Conclusion

This paper makes two contributions. First, we show that liquidation cascades can be detected in real time from public order book data, bypassing the opacity of exchange-reported feeds (Lunde, 2024). Our framework uses freely accessible WebSocket channels and labels events via the trading system’s exit logic, providing a replicable methodology.

Second, and more importantly, we document substantial cross-coin heterogeneity in liquidation cascade dynamics among three meme coin perpetual futures that trade on the same exchange and share the same liquidation engine. The large-cap, deep-book coin (PEPE) exhibits a fundamentally different cascade profile from the two small-cap, thin-book coins (BOME and FLOKI): lower positive cascade rates (12.5% vs. 19.4% and 18.2%), stronger OBI–Bias coupling ($\rho = -0.23$ vs. +0.14), and lower trailing-stop activation (3.8% vs. 10.2%). The PEPE–BOME and PEPE–FLOKI differences are statistically significant at $p < 0.001$ under bootstrap.

The OBI–Bias sign reversal from PEPE’s negative coupling ($\rho = -0.23$) to BOME’s positive coupling ($\rho = +0.14$) is the paper’s most striking finding. In PEPE, deepening

sell pressure predicts price declines—the canonical microstructure channel. In BOME, the opposite holds: extreme sell-side imbalance predicts price rebounds, consistent with rapid contrarian buying in a thin, fast-moving market. FLOKI ($\rho = -0.14$) sits between the two regimes. This implies that microstructure findings from one asset class—or even from one blockchain—may fail to generalize to assets with fundamentally different market depth and speed.

4.1 Cross-Exchange Validation

To assess whether our OKX-based findings are exchange-specific, we compare bid-ask spreads and mid-price volatility across Binance and Bybit using REST tick data collected at 5-second intervals (39,600 observations per coin per exchange). Table 3 reports the results. BOME and FLOKI spreads significantly exceed PEPE spreads on both Binance and Bybit (Mann–Whitney, all $p < 0.001$). Mid-price volatility is uniformly higher for BOME than PEPE across all three venues. These results confirm that the cross-coin heterogeneity in market depth and volatility is not an OKX artifact—the ranking of small-cap meme coins above large-cap PEPE holds across exchanges, consistent with the interpretation that market capitalization and order book depth, rather than exchange-specific matching-engine design, drive the observed heterogeneity.

4.2 Limitations

Our study has several limitations. First, data collection spans 3.5 days (June 2–5, 2026). While this window is short in calendar time, it forms a natural experiment that compresses calm conditions (June 2–3) and an extreme sell-off (June 4–5, BTC down 13.5%, over \$1.5 billion in liquidations) into a single contiguous sample. Additional market regimes (e.g., sustained bull trends) would further strengthen generalizability. Second, while cross-exchange validation confirms that spread and volatility patterns generalize, our cascade detection framework itself has been deployed only on OKX; replication of the full four-path trigger system on Binance and Bybit WebSocket data would further strengthen generalizability. Third, with only three coins we cannot disentangle the struc-

tural drivers—market capitalization, blockchain architecture, and investor composition all covary. The BOME–FLOKI similarity suggests market depth as the binding constraint, but a larger cross-section is required for formal identification. Fourth, our detection framework is validated against its own labeling logic rather than an external ground truth. Cross-validation against REST-reported liquidation data and on-chain liquidation records from decentralized perpetual exchanges such as Hyperliquid, where liquidations are publicly verifiable, represents a promising direction for future work (Gudgeon et al., 2020).

Notwithstanding these limitations, our findings carry practical implications. For exchanges, the documented heterogeneity suggests that uniform risk parameters—margin requirements, position limits, and liquidation thresholds—may be mis-calibrated when applied across coins with fundamentally different market microstructure. A meme coin with Solana’s ~ 0.4 -second blocks and BOME’s thin order books bears a different cascade risk profile than one with Ethereum’s ~ 12 -second blocks and PEPE’s deep liquidity, even if both are classified as “meme coins” by the exchange. For researchers, our results caution against the tacit assumption that findings from cryptocurrency X generalize to cryptocurrency Y . The microstructure matters—and it varies with the chain.

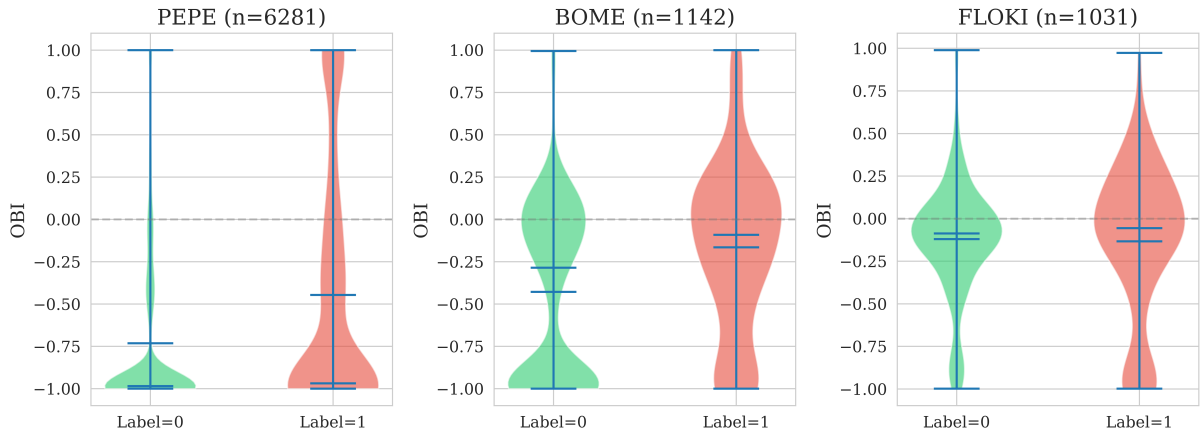


Figure 1: OBI Distribution by Coin and Liquidation Outcome

Note: Violin plots show OBI distributions split by cascade outcome. PEPE is strongly left-skewed (persistent sell-side pressure); BOME shows moderate left skew; FLOKI is near symmetric.

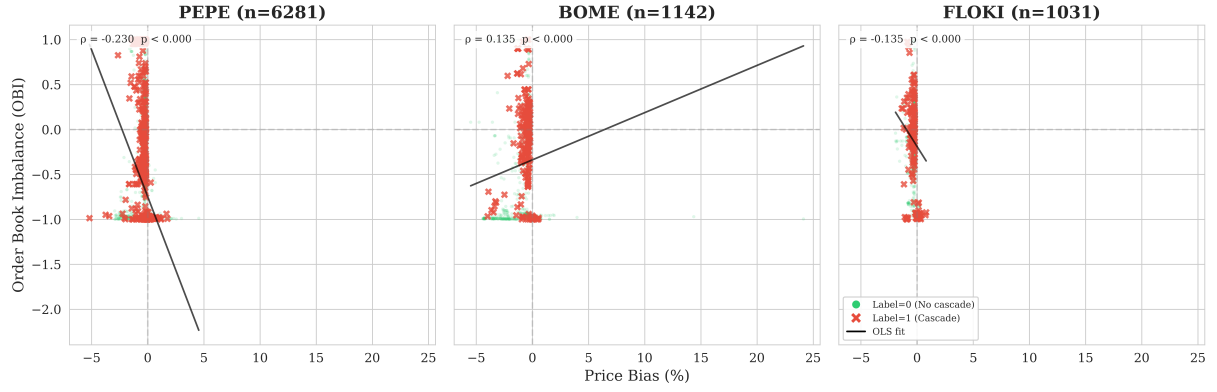


Figure 2: OBI-Bias Coupling Across Coins

Note: OBI versus Price Bias with OLS regression lines. PEPE: $\rho = -0.230$ ($p < 0.001$); BOME: $\rho = +0.135$ ($p < 0.001$); FLOKI: $\rho = -0.135$ ($p < 0.001$). All pairwise differences significant (Fisher z -test: $z = 11.47, 2.93, 6.30$; all $p < 0.01$).

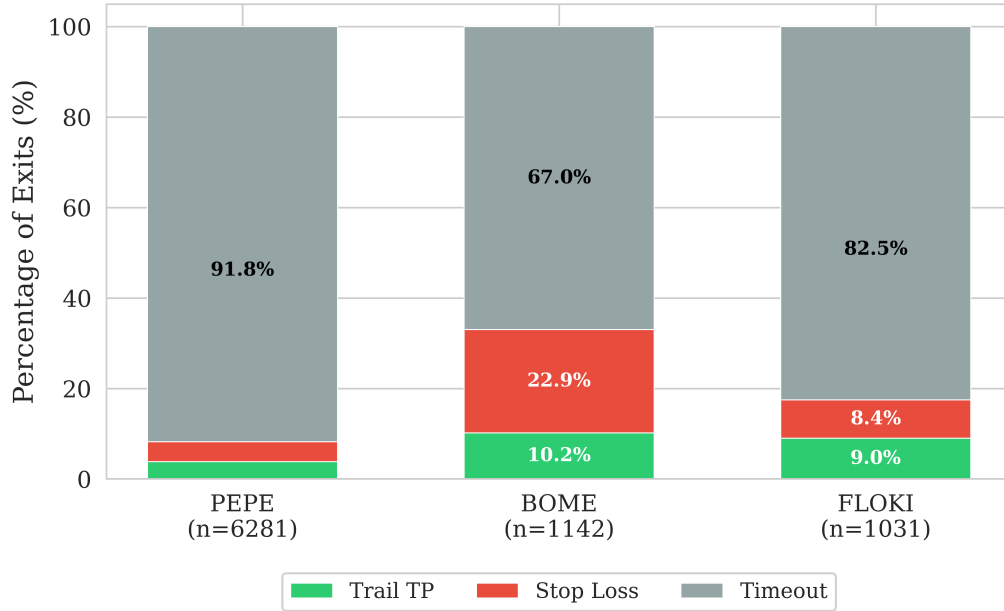


Figure 3: Exit Reason Composition by Coin

Note: Stacked bars show the percentage distribution of exit reasons for each coin. FLOKI and BOME exhibit higher trailing-stop activation rates (9.0% and 10.2%) than PEPE (3.8%). BOME has the highest stop-loss rate (22.9%); PEPE's events predominantly time out (91.8%).

Declaration of Generative AI and AI-Assisted Technologies

During the preparation of this work, the author used Claude (Anthropic) for language polishing, reference formatting, and LaTeX typesetting. The author takes full responsibility for all research design, data collection, statistical analysis, scientific conclusions, and the content of the published article.

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